

GroupTrack User Manual

Version 2.5 Addendum

Off-Road Convoy Tracking for Android

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This addendum documents new and updated features in GroupTrack V2.5. It supplements the V2.4 User Manual. Chapter numbers reference the V2.4 manual structure. Updated chapters replace their V2.4 counterparts entirely.

Chapter 6. Work With Artifacts (replaces Display Panel)

V2.5 replaces the simple Display Panel with the Work with Artifacts panel, a comprehensive artifact management interface. The panel provides toggle controls for all four artifact types plus access to editing and import functions.

6.1 Opening Work With Artifacts

On the Planning Map, tap the WORK WITH ARTIFACTS button in the display area. The panel expands to show all artifact types with their counts and toggle controls.

On the Convoy Map, the same panel is available but in display-only mode. Editing and import functions are not available during live convoy operations.

6.2 Artifact Type Grid

The panel shows four artifact types in a grid layout. Each row displays the artifact type name, count, and a display toggle switch.

Trails

Toggle ON to display trail overlays from the spatial database. Trails are loaded dynamically based on your current map viewport. As you pan and zoom, trails update automatically. Only trails visible in your current view are loaded, ensuring smooth performance even with large trail datasets.

Trails appear as cyan (#00FFFF) solid lines with weight 4 on the map. Trail color varies by classification: blue for OHV/road-concurrent, orange for hiking/biking, yellow for hiking-only, purple for biking-only, and cyan for general/paved trails.

Trails load at zoom level 8 and above. At wider zoom levels, trails are not displayed to maintain map readability. Zoom in to see trail overlays.

Tracks

Toggle ON to display your recorded GPS tracks from the spatial database. Like trails, tracks are loaded dynamically based on your viewport. All tracks from your my_tracks/ library are automatically synced to the spatial database on app launch.

Tracks appear as neon green (#39FF14) dashed lines with weight 3 on the map. The dashed pattern distinguishes tracks from trails at a glance. Tracks persist on the map as you pan and zoom.

Waypoints (V2.5 In Progress)

Toggle for user-created waypoints. 10 standard waypoint types: trailhead, fuel, gate, hazard, scenic viewpoint, water source, camp, parking, rally point, and other. Each type has a distinct color and icon on the map.

Routes (V2.5 In Progress)

Toggle for planned routes including drawn routes, snap-to-trail routes, and imported route files.

6.3 Import Buttons

Two import buttons appear at the bottom of the Work with Artifacts panel on the Planning Map:

IMPORT TRAILS

Opens the trail source management screen where you can download trail databases from public land management agencies for your region. Trail data is stored in the spatial database for fast viewport-based display.

IMPORT ARTIFACTS

Opens a file picker to import GPX files from your Downloads folder. The import process automatically detects tracks, routes, and waypoints within the GPX file and inserts them into the appropriate spatial database tables. Tracks are also saved as GPX files in your my_tracks/ directory.

6.4 Trails and Tracks Together

Both Trails and Tracks can be displayed simultaneously. Trails render as solid cyan lines and tracks as dashed neon green lines, making them visually distinct even when overlapping. Toggle each independently using the display switches.

Trail and track data is queried from an on-device spatial database using bounding box intersection. Only artifacts within your current map viewport are loaded. This architecture supports thousands of trails and hundreds of tracks without performance degradation.

Chapter 9. Track Display (Updated for V2.5)

V2.5 introduces spatial database-backed track display with viewport-aware lazy loading. Tracks are automatically synced from your GPX files into the spatial database and displayed based on your current map viewport.

9.1 Automatic Track Sync

When you open the Planning Map, GroupTrack automatically scans your `my_tracks/` directory and syncs any new GPX files into the spatial database. This one-time sync extracts track geometry and computes bounding boxes for fast viewport queries. Files already in the database are skipped.

The sync process runs in the background and does not block map interaction. New tracks appear on the map within seconds of the sync completing.

9.2 Viewport-Based Display

Unlike V2.4 which loaded all tracks sequentially from disk, V2.5 queries the spatial database for tracks whose bounding boxes intersect your current map viewport. This means only visible tracks are loaded and rendered, providing instant response even with large track libraries.

As you pan and zoom the map, the track query updates automatically with a 400-millisecond debounce to prevent excessive queries during rapid scrolling.

9.3 Track Appearance

Tracks display as dashed neon green lines (`#39FF14`) with a dash pattern of 10 pixels on, 5 pixels off, and a line weight of 3. This dashed pattern clearly distinguishes tracks from trails (solid lines) on the map.

Tapping a track line shows a popup with the track name. Track names are derived from the GPX filename.

9.4 Track Colors

All tracks displayed via the spatial database toggle appear in the standard neon green (`#39FF14`) dashed pattern. Individual track loading with color rotation is available through the TRACKS button on the Planning Map header (unchanged from V2.4).

Chapter 9A. Trail Display (New in V2.5)

V2.5 introduces spatial database-backed trail display. Trail data from public land management agencies is stored in an on-device GeoPackage/SQLite database and displayed dynamically based on your map viewport.

9A.1 Trail Data Coverage

The current release includes trail data for southern Utah and northern Arizona, covering the St. George to Grand Canyon corridor. This includes OHV trails, multi-use trails, hiking paths, and paved shared-use paths from USFS, BLM, and state land management agencies.

Additional state and regional trail datasets will be available for download in future updates through the Trail Source Management screen.

9A.2 Trail Appearance

Trails display as solid cyan lines (#00FFFF) with a line weight of 4. Trail color varies by CartoCode classification:

Blue (#00AAFF): OHV and road-concurrent trails (CartoCode 4) with weight 5 for emphasis.

Orange (#FF8800): Hiking and biking trails (CartoCode 2).

Yellow (#FFCC00): Hiking-only trails (CartoCode 1).

Purple (#AA44FF): Biking-only trails (CartoCode 5).

Cyan (#00FFFF): Default for unclassified or paved shared-use trails.

9A.3 Viewport-Based Loading

Trails load dynamically as you pan and zoom the map. The spatial database uses bounding box indexing to rapidly identify trails visible in your current viewport. Trails appear at zoom level 8 and above.

At wide zoom levels (below 8), trails are not displayed to maintain map readability and prevent performance issues from loading thousands of trail segments simultaneously.

9A.4 Trail Properties

Each trail in the database includes properties from the originating land management agency: primary name, surface type (paved, unpaved, gravel), designated uses (motor, hike, horse, bike), motorized access status, trail class, steward organization, and county. These properties are available in the trail detail panel when tapping a trail on the map (coming in a future update).

Chapter 10. Download Queue (Updated for V2.5)

V2.5 enhances the download queue with management controls including hold, resume, cancel individual entries, and cancel all operations.

10.1 Queue Management Controls

The WORK WITH QUEUES button on both the Convoy Map and Planning Map opens the queue management panel. The panel shows active, queued, and completed download entries.

Hold Queue

Pause all queue processing. Active downloads complete their current tile but no new tiles are started and no queued items begin processing. Use this when you need bandwidth for other activities.

Resume Queue

Resume queue processing after a hold. Paused downloads continue from where they stopped and queued items begin processing.

Cancel Entry

Cancel an individual download entry. The entry is removed from the queue and any in-progress tiles for that entry stop downloading. Downloaded tiles for the partially completed entry are preserved on disk.

Cancel All

Stop all active downloads and clear the entire queue. All in-progress downloads stop immediately. Previously downloaded tiles are preserved.

Queue management controls are fully functional in the backend. The queue panel UI is being wired to live data in the next update.

Appendix C. V2.5 Feature Status

This appendix replaces V2.4 Appendix C with the current status of V2.5 features.

C.1 Completed Features

✓	Trail lazy load from spatial database with viewport-based queries
✓	Track lazy load from spatial database with GPX sync
✓	Work with Artifacts panel (display toggles, import buttons)
✓	Trail display: cyan solid lines, CartoCode-based coloring
✓	Track display: neon green dashed lines from spatial DB
✓	Automatic GPX-to-spatial-DB sync on app launch
✓	Tile download system fix (CalTopo removed, asset-only reads, slot filter)
✓	Queue management backend (hold, resume, cancel, cancel all)
✓	Expanded tile source catalog (16 sources across 6 categories)
✓	API key management for premium tile sources
✓	Schema v2 migration (track geometry + bounding box columns)
✓	Spatial database architecture (trails, tracks, waypoints, routes tables)
✓	Trail and track viewport persistence on pan/zoom
✓	Minimum zoom level 8 for artifact display

C.2 In Progress

⚠	Queue panel UI wired to live data
⚠	GPX import for waypoints and routes
⚠	Track CRUD sync to spatial database on create/edit/delete
⚠	Trail/track display on convoy map
⚠	Artifact detail panels (tap trail/track for properties)
⚠	Waypoint creation (long-press on map)
⚠	Route creation (draw/snap/convert)
⚠	Trail source management (download regional datasets)

C.3 Planned for V2.5 (Not Yet Started)

❑	Waypoint lazy load from spatial database
❑	Route lazy load from spatial database
❑	Alias system (user-assigned names for agency trails)
❑	Color trails by motorized_allowed classification
❑	Remove downloaded tile areas (draw-to-delete)
❑	Downloads in progress overlay (pink/red shading)
❑	Trip odometer / mileage display
❑	Display panel position persistence

C.4 Deferred to V2.6 / V3.0

☐	iPad / iOS companion app (V2.6)
☐	Map heading-up rotation (V2.6)
☐	National trail model with multi-state downloads (V2.6)
☐	V3.0 dashboard rewrite with cloud features
☐	Cloud sync of local artifacts to web account (V3.0 premium)
☐	Database backup and recovery (V3.0 premium)
☐	Organization services tier (V3.0)